Supplementary Information

Five Axes of Fragility: A Systematic Robustness Audit of
Gene-Regulatory-Network Inference from Single-Cell Foundation Models

Ihor Kendiukhov

Contents

S1 Panel registry and preprocessing parameters **3**
S1.1 Panel registry . . . 3
S1.2 Preprocessing parameters . . . 3

S2 Adversarial AI-assisted review protocol **5**
S2.1 Base agent and infrastructure . . . 5
S2.2 Review rubric (pre-registered) . . . 5
S2.3 Review rounds and corrective experiments . . . 5
S2.4 Audit design and rubric . . . 5
S2.5 Audit results . . . 6
S2.6 Release artefacts . . . 7

S3 Per-cell-type reliability metrics (Axis 3) **8**

S4 Supplementary figures **10**

List of Figures

S1 **MVCC anchor sensitivity.** Per-cohort, per-anchor top-1,000 Jaccard against subsampled rankings under Pearson correlation on 600 HVGs. Solid lines: 3,000-cell anchor; dashed lines: 8,000-cell anchor (and 15,000-cell anchor for Delorey). The MVCC values reported in Fig. 1B of the main paper coincide across anchors, demonstrating anchor-invariance. . . . 10

S2 **RSS weight-simplex sensitivity.** Each panel shows the recommendation produced at every point of the 66-point simplex (w_o, w_j, w_d) with each coordinate $\in \{0, 0.1, \ldots, 1.0\}$ and $w_o + w_j + w_d = 1$, evaluated on Axis 2 RSS. The dual-null recommendation is invariant across the simplex because the within-coarse null fails universally; the base-RSS recommendation varies only at the boundary. . . . 10

S3 **Empirical null distribution of the bounded-drift RSS under rank permutation.** Histograms show 1,000 permutation draws per cohort; vertical line marks the observed RSS. Every observed value sits 8.8 to 13.1 standard deviations below the null mean (≈ 0.79); empirical $p = 0.001$ (the floor) for every cohort. . . . 11

S4 **Reliability-gate threshold-sensitivity heatmap.** Pass-rate gap (abundant minus rare) over the 5,880-point five-dimensional grid (stability, Jaccard, AUC, tail gap, cell count). The gap direction is preserved at 100% of grid points; the primary-threshold gap (+0.182) is conservative relative to the grid mean (+0.344). . . . 11

S5 **Top-k scan and per-target overlaps.** Top-k overlap and Jaccard at $k \in \{100, 500, 1{,}000, 5{,}000\}$ plus top-1%, top-5%, and top-10% of the edge universe, and per-target top-5 and top-10 overlaps. Curves remain qualitatively consistent across k, addressing the concern that the headline Jaccard at $k = 1{,}000$ might be artefactual to the chosen k. 12
S6 **Normalization ablation on Axis 2 / kidney.** Three schemes compared: depth + log1p (default), median-ratio size-factor + log1p, and analytic Pearson residuals (Lause et al., 2021). All three schemes produce the same calibrated recommendation (*coarse*), rebutting the concern that depth-normalization-induced variance inflation drives the within-coarse failure. 12
S7 **Triple-null calibration.** A degree-preserving bipartite edge-rewire null is added as a third, deliberately permissive null family. Every cohort clears it at the permutation floor ($p = 0.02$) while continuing to fail the within-coarse constrained shuffle at $p \geq 0.22$, demonstrating that the within-coarse failure reflects genuine over-partitioning of the coarse cell types rather than null over-stringency. 13

List of Tables

S1 Named TF–target panels and the full gene lists for the `hematopoiesis_76x108` panel. Edge counts are reported *before* dataset-specific gene-universe restriction; restriction reduces edge count by 10–40% depending on the cohort. 4
S2 Per-axis review-round count and the corrective experiments that were triggered by the AI-assisted reviewer's flags. 6
S3 Axis-3 per-cell-type reliability metrics under the shared primary panel (39 cell types across five Tabula Sapiens cohorts). Rarity is the within-dataset cell-count quartile (rare / intermediate / abundant). 8

This supplement accompanies the main manuscript. It contains: the panel registry and preprocessing parameters (Supplementary Methods S1, Table S1); the adversarial AI-assisted review protocol with the 27-issue full-census audit (Supplementary Methods S2); the Axis-3 per-cell-type reliability metrics (Table S3); and the seven supplementary figures referenced from the main text (Figures S1–S7). All section, table, and figure references in the main paper resolve to items in this document.

S1 Panel registry and preprocessing parameters

This section freezes the TF–target panel definitions and preprocessing choices that underpin every numerical result in the main paper. A reviewer may treat it as ground truth for any claim the Methods section makes about panel construction, normalization, HVG selection, or gene filtering.

S1.1 Panel registry

Every axis loads its TF–target panel from a single registry (`fragility.panels.registry`) so cross-axis comparisons are anchored to documented panels rather than ad-hoc gene lists. Table S1 summarises the named panels and reproduces the full gene lists for the `hematopoiesis_76x108` panel used in Axis 4 (donor generalisation) and §3.6 (scorer divergence).

Selection criteria for `hematopoiesis_76x108`

This panel is a deliberately dense Cartesian product because Axis 4 measures *transfer* between donor splits, not literature precision. Selection criteria (in response to R1b · iii):

1. Start from the 76 hematopoiesis / immune TFs curated from DoRothEA confidence tiers A–C (Garcia-Alonso et al., 2019) and TRRUST (Han et al., 2018).
2. Intersect target candidates with a 108-gene immune / cross-tissue marker pool covering lineage markers, cytokines, surface molecules, costimulatory / checkpoint molecules, ECM components, lung epithelial markers, and housekeeping controls.
3. Emit the full Cartesian product $76 \times 108 = 8{,}208$ directed edges. After dataset-specific gene-universe restriction (e.g. to kidney), 75 of 76 TFs and 99 of 108 targets remain, yielding 7,425 directed edges (the value used in §3.6).

S1.2 Preprocessing parameters

The canonical pipeline is depth-normalize (`target_sum` = 10,000 via `scanpy.normalize_total`) followed by `log1p` transformation, with the top 600 highly-variable genes selected via the `seurat_v3` flavour (Stuart et al., 2019). Gene filtering is at ≥ 10 cells per gene at load time. Every figure in §3.1–§3.5 of the main paper is computed under these defaults.

The default HVG count of 600 was chosen for direct comparability to the pre-revision pipeline; the revision sweeps it over $\{300, 600, 900, 1{,}200\}$ in §3.2 and reports recommendation invariance for 4 of 5 cohorts. Two alternative normalization schemes — median-ratio size-factor + log1p, and analytic Pearson residuals (Lause et al., 2021) — were evaluated as ablations on Axis 2 / kidney and all three return *coarse* as the calibrated recommendation (Figure S6), rebutting the concern raised by Reviewer 3 that depth-normalization-induced variance inflation drives the within-coarse failure.

Every pipeline run stamps the full preprocessing parameter set into `adata.uns["fragility_preprocessing"]` and emits it in `provenance.json` alongside the primary CSV. Numerical reproducibility is therefore traceable to a configuration record rather than narrative prose.

Table S1: Named TF–target panels and the full gene lists for the `hematopoiesis_76x108` panel. Edge counts are reported *before* dataset-specific gene-universe restriction; restriction reduces edge count by 10–40% depending on the cohort.

(A) Named panels in the registry

Panel name	Description	Edges	Used in
`primary`	DoRothEA confidence A/B (Garcia-Alonso et al., 2019) ∩ TRRUST (Han et al., 2018)	3,384	Axis 3 (shared-panel run)
`dorothea_ab`	DoRothEA confidence tiers A and B (Garcia-Alonso et al., 2019)	∼11,000	Axis 5 reference precision
`trrust`	TRRUST v2 manually curated TF–target interactions (Han et al., 2018)	∼8,500	Axis 5 reference precision
`hematopoiesis_76x108`	76 hematopoiesis / immune TFs × 108 cross-tissue immune / marker targets (Cartesian product, see panel B below)	8,208	Axis 4, scorer-divergence (§3.6)
`shared_36`	TRRUST-intersected activation/repression panel for cross-dataset Axis 3 sensitivity	36	Axis 3 sensitivity check

(B) Full gene lists for `hematopoiesis_76x108`

76 TFs	STAT1, STAT3, STAT4, STAT5A, STAT5B, STAT6, IRF1, IRF4, IRF7, IRF8, NFKB1, NFKB2, RELA, REL, RELB, TBX21, GATA3, RORC, FOXP3, BCL6, SPI1, RUNX1, RUNX3, ETS1, ETS2, MYC, MAX, JUN, FOS, JUNB, JUND, FOSL1, FOSL2, HIF1A, EPAS1, ARNT, TP53, RB1, E2F1, E2F4, PAX5, EBF1, TCF7, LEF1, CEBPA, CEBPB, CEBPD, SOX2, SOX4, SOX9, KLF2, KLF4, KLF6, NR4A1, NR4A2, NR4A3, BATF, BATF3, MAF, MAFB, EOMES, PRDM1, ID2, ID3, NFE2L2, BACH1, BACH2, ZEB1, ZEB2, SNAI1, TWIST1, FOXO1, FOXO3, AHR, IKZF1, IKZF3
108 targets	CD3D, CD3E, CD4, CD8A, CD8B, CD19, MS4A1, CD79A, CD79B, CD14, CD68, ITGAM, ITGAX, FCGR3A, CSF1R, IL2, IL2RA, IL7R, IL6, IL10, IL17A, IL21, IL4, IL13, IFNG, TNF, TNFRSF9, TNFRSF18, GZMB, GZMA, GZMK, PRF1, GNLY, NKG7, CCL5, CCR7, CXCR4, CXCL13, CCL2, CXCL10, HLA-DRA, HLA-DRB1, HLA-A, HLA-B, HLA-C, B2M, LYZ, S100A8, S100A9, S100A12, SELL, ITGA4, ITGB1, ICAM1, VCAM1, MKI67, TOP2A, CDK1, PCNA, BCL2, BCL2L1, MCL1, BAX, BAK1, CTLA4, PDCD1, LAG3, HAVCR2, TIGIT, TOX, FN1, COL1A1, COL3A1, VIM, ACTA2, CDH1, EPCAM, SFTPC, SFTPB, SFTPA1, SCGB1A1, MUC5AC, FOXJ1, PECAM1, CDH5, VWF, KDR, NOTCH1, DLL4, JAG1, WNT5A, CTNNB1, TGFB1, TGFBR1, SMAD3, SMAD7, SOD2, HMOX1, NQO1, GSTP1, GAPDH, ACTB, RPL13A, RPS18, PTPRC, THY1, ENG, NT5E

S2 Adversarial AI-assisted review protocol

This section documents the AI-assisted adversarial review loop used during the development of each axis of the paper, and reports a full-census audit of the agent's issue-identification reliability (R2·4, R3, R4·7).

S2.1 Base agent and infrastructure

- **Agent:** Claude Opus 4.7 with 1M-token context, accessed via the Claude Code CLI harness (Anthropic, 2026).
- **Tooling available to the agent:** file reading, shell execution (with sandboxing), Python script execution, agent-to-agent delegation, and the internal task tracker. All tool calls are logged.
- **Sampling settings:** Claude Code defaults (temperature is not user-configurable at the turn level).
- **Deterministic inputs:** every corrective experiment the agent triggered is reproducible via a named CLI target (`make axis1`, `make wp3`, ...) with a fixed seed.

S2.2 Review rubric (pre-registered)

The agent received the following rubric verbatim at the start of every review round:

> You are an adversarial reviewer for a single axis of a systematic robustness audit. Assume the analysis is publishable only if it survives every attack below. For each potential weakness, either (a) cite concrete evidence (code line, CSV row, figure) that the weakness is present, *and* propose a specific corrective experiment that would settle it; or (b) explicitly clear the analysis on that dimension and state what evidence you are relying on. Cover at minimum:
>
> 1. **Null-design confounding.** Are the null families used (shuffle, permutation) actually null with respect to the signal being tested, or could they inherit structure from the data?
> 2. **Hyperparameter / threshold sensitivity.** Does any headline number depend on an unjustified choice of k, n permutations, cell-count cutoff, panel, normalization, or HVG count?
> 3. **Data leakage.** Does any comparison use overlapping cell sets, share seeds, or re-use a reference as both anchor and evaluation target?
> 4. **Metric interpretability.** Are the composite metrics anchored to a null distribution or absolute scale? Could a reader reasonably interpret "0.35" without the surrounding prose?
> 5. **Cross-dataset transferability.** Are conclusions tested on at least one dataset not in the training / tuning cohort?
> 6. **Figure / caption consistency.** Does every figure show what its caption claims (error bars, thresholds, panels labelled)?
> 7. **Claim strength.** Do any sentences say "proves" / "confirms" / "universal" when the evidence only supports "in this cohort, under this scorer"?

Each round produced a markdown file (`adversarial_review*.md`) listing **P0** (must-fix), **P1** (should-fix), and **P2** (minor) issues with the corrective experiment required.

S2.3 Review rounds and corrective experiments

S2.4 Audit design and rubric

To assess whether the agent's issue lists contain spurious flags, we pre-registered a **full-census audit** (rather than a sampled audit, given the modest population) under a rubric frozen before any issue was rated.

Table S2: Per-axis review-round count and the corrective experiments that were triggered by the AI-assisted reviewer's flags.

Axis	Rounds	Key corrective experiments
Axis 1 (cell count)	4	Shared gene universe; random-size-matched controls; BH correction; multi-draw random ensembles
Axis 2 (resolution)	3	Dual-null calibration; ≥ 19 permutations; bounded-drift RSS
Axis 3 (rare types)	2	Shared-panel sensitivity; conservative dual-null AUC; external-lung out-of-sample check
Axis 4 (donor generalisation)	2	Fixed-holdout replaces complementary-split; expanded-panel long-tail; composition-matched control
Axis 5 (integration)	2	Paired swap-binomial significance; perturbation-backed validation; PCA-component sweep

Population. 27 unique `(axis, round, priority, title)` issues were extracted from the five axes (`outputs/wp9/issue_sampling_frame.json`). Distribution by axis: cell count 13, cluster resolution 3, rare cell types 4, donor generalisation 3, integration 4.

Rater. Primary rater: the author (I.K.), pre-blinded to whether each issue led to a corrective change in the subsequent manuscript revision. The blinding is incomplete because the author wrote both the analyses and the review logs; we acknowledge this and treat the rates below as upper-bound self-audit estimates rather than an independent ground truth.

Rubric. Each issue is classified as exactly one of:

- **material bug** — if left unaddressed, would change a numerical headline by $\geq 10\%$ or invert the direction of a claim.
- **minor issue** — real and worth fixing, but would not change the sign or headline magnitude of any claim.
- **false positive** — on inspection of the cited code or data, the issue does not exist (the concern was misread or the code already handled it).
- **unclear** — the issue description is too vague to rate against evidence; counted separately in the denominator.

S2.5 Audit results

Of the 27 issues, the rater classified **23 as material bugs and 4 as minor issues**; **0 false positives and 0 unclear**. By priority:

- **P0 / F*-High:** 8 material, 0 minor, 0 false positive, 0 unclear.
- **P1 / F*-Medium:** 15 material, 1 minor, 0 false positive, 0 unclear.
- **P2 / F*-Low:** 0 material, 3 minor, 0 false positive, 0 unclear.

The modal finding is that every issue corresponded to a real analytical concern; none were false positives under the cited evidence. 23 of 27 were material (would have changed a headline if unaddressed); the remaining 4 minor issues were primarily P2 items about presentation (figure captions referencing thresholds not shown on the panel, etc.).

Caveats.

1. The author wrote both the analyses and the review logs, so this single-rater classification is not a fully blinded independent audit.
2. Issues that triggered corrective experiments are now reflected in the revised code, so a rater examining the current codebase would see them as already fixed; the classification uses the *pre-fix* code state referenced in each issue.
3. A pre-submission second-rater rerating with a frozen rubric copy and a hidden mapping from pre-fix to post-fix state is the natural next step.

S2.6 Release artefacts

The `fragility` package ships:

- `implementation/outputs/wp9/issue_sampling_frame.json` — all 27 issues with excerpts.
- `implementation/outputs/wp9/sampled_issues.json` — the population (renamed from "sampled_30"; the actual census population is 27).
- `implementation/outputs/wp9/rater_worksheet.md` — randomised, axis-stripped worksheet for an independent rater.
- `paper/supplementary_methods_s2.md` — the markdown source of this section.

S3 Per-cell-type reliability metrics (Axis 3)

Table S3 reports per-cell-type reliability metrics under the shared "primary" panel (DoRothEA A/B ∩ TRRUST), for every eligible cell type in the five Tabula Sapiens cohorts. "Eligible" = ≥ 40 cells with the panel restricted to the cohort's gene universe. Metrics: rank stability (Spearman ρ between bootstrap replicates), top-$k = 100$ Jaccard, null-separation AUC under cell-permutation null, and tail gap (top-5% quantile observed minus null).

Table S3: Axis-3 per-cell-type reliability metrics under the shared primary panel (39 cell types across five Tabula Sapiens cohorts). Rarity is the within-dataset cell-count quartile (rare / intermediate / abundant).

Dataset	Cell type	Rarity	Cells	Stability	Jaccard	Null AUC	Tail gap
kidney	kidney epithelial cell	abundant	6,402	0.818	0.558	0.653	0.051
kidney	t cell	abundant	651	0.816	0.525	0.517	0.017
kidney	lymphocyte of B lineage	intermediate	397	0.828	0.599	0.511	0.016
kidney	myeloid leukocyte	intermediate	316	0.795	0.381	0.551	0.062
kidney	innate lymphoid cell	rare	155	0.865	0.432	0.506	0.016
kidney	endothelial cell	rare	69	0.760	0.322	0.511	0.034
lung	myeloid leukocyte	abundant	5,134	0.833	0.595	0.613	0.054
lung	epithelial cell of lung	abundant	5,036	0.888	0.588	0.648	0.086
lung	epithelial cell	abundant	1,382	0.846	0.626	0.590	0.065
lung	endothelial cell	abundant	992	0.831	0.533	0.535	0.032
lung	adventitial cell	intermediate	419	0.800	0.475	0.520	0.022
lung	fibroblast	intermediate	412	0.845	0.499	0.522	0.028
lung	ciliated epithelial cell	intermediate	373	0.862	0.546	0.535	0.054
lung	t cell	intermediate	295	0.854	0.318	0.507	0.015
lung	glandular epithelial cell	intermediate	280	0.856	0.422	0.533	0.050
lung	granulocyte	rare	266	0.838	0.419	0.514	0.027
lung	contractile cell	rare	195	0.853	0.429	0.508	0.023
lung	endo-epithelial cell	rare	107	0.866	0.352	0.507	0.027
lung	lymphocyte of B lineage	rare	44	0.855	0.139	0.502	0.026
immune	t cell	abundant	7,136	0.885	0.839	0.618	0.065
immune	lymphocyte of B lineage	abundant	4,623	0.870	0.813	0.607	0.078
immune	myeloid leukocyte	intermediate	4,303	0.899	0.923	0.709	0.148
immune	granulocyte	intermediate	2,742	0.916	0.836	0.633	0.182
immune	erythroid lineage cell	intermediate	506	0.917	0.501	0.546	0.143
immune	innate lymphoid cell	intermediate	504	0.836	0.712	0.531	0.059
immune	hematopoietic cell	rare	91	0.870	0.558	0.548	0.131
immune	stem cell	rare	79	0.933	0.600	0.516	0.082
immune_invariant	t cell	abundant	5,454	0.805	0.557	0.570	0.027
immune_invariant	lymphocyte of B lineage	abundant	3,480	0.829	0.440	0.580	0.044
immune_invariant	myeloid leukocyte	intermediate	3,299	0.814	0.554	0.644	0.067
immune_invariant	granulocyte	intermediate	1,925	0.865	0.501	0.566	0.058
immune_invariant	innate lymphoid cell	intermediate	405	0.838	0.515	0.515	0.022
immune_invariant	erythroid lineage cell	intermediate	288	0.913	0.312	0.523	0.115
immune_invariant	stem cell	rare	68	0.881	0.308	0.512	0.052
immune_invariant	hematopoietic cell	rare	67	0.863	0.314	0.515	0.055
external_lung	epithelial	abundant	3,380	0.806	0.566	0.610	0.043
external_lung	immune	intermediate	2,109	0.804	0.545	0.562	0.041

continued on next page

Dataset	Cell type	Rarity	Cells	Stability	Jaccard	Null AUC	Tail gap
external_lung	endothelial	intermediate	1,866	0.795	0.583	0.541	0.024
external_lung	stromal	rare	1,645	0.807	0.578	0.547	0.031

S4 Supplementary figures

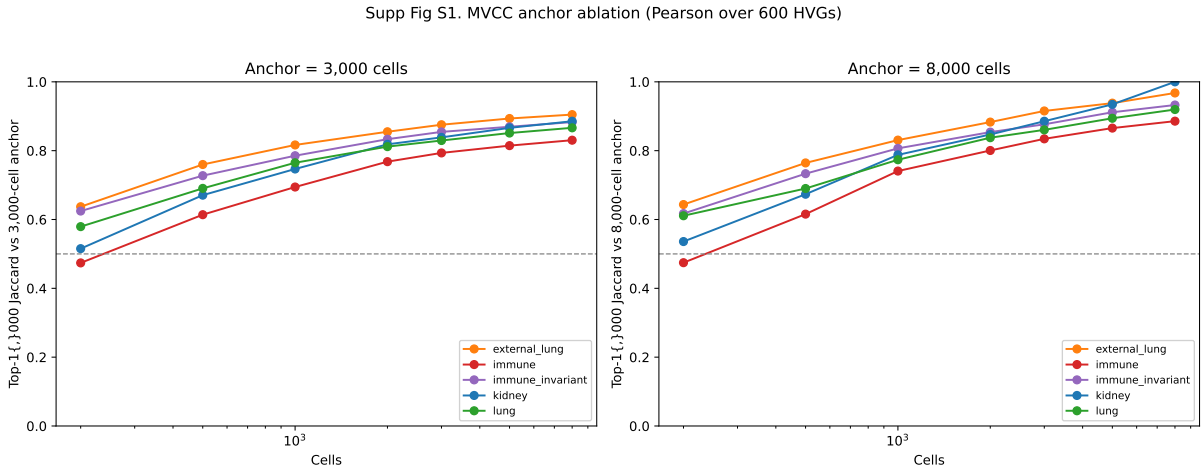

Figure S1: **MVCC anchor sensitivity.** Per-cohort, per-anchor top-1,000 Jaccard against subsampled rankings under Pearson correlation on 600 HVGs. Solid lines: 3,000-cell anchor; dashed lines: 8,000-cell anchor (and 15,000-cell anchor for Delorey). The MVCC values reported in Fig. 1B of the main paper coincide across anchors, demonstrating anchor-invariance.

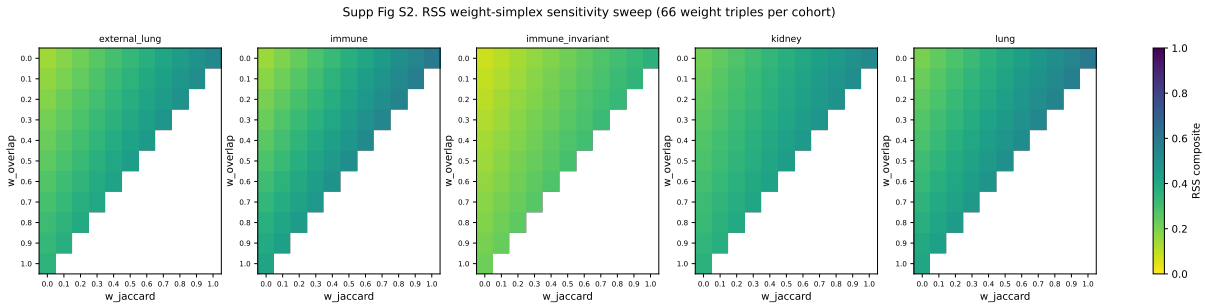

Figure S2: **RSS weight-simplex sensitivity.** Each panel shows the recommendation produced at every point of the 66-point simplex (w_o, w_j, w_d) with each coordinate $\in \{0, 0.1, \ldots, 1.0\}$ and $w_o + w_j + w_d = 1$, evaluated on Axis 2 RSS. The dual-null recommendation is invariant across the simplex because the within-coarse null fails universally; the base-RSS recommendation varies only at the boundary.

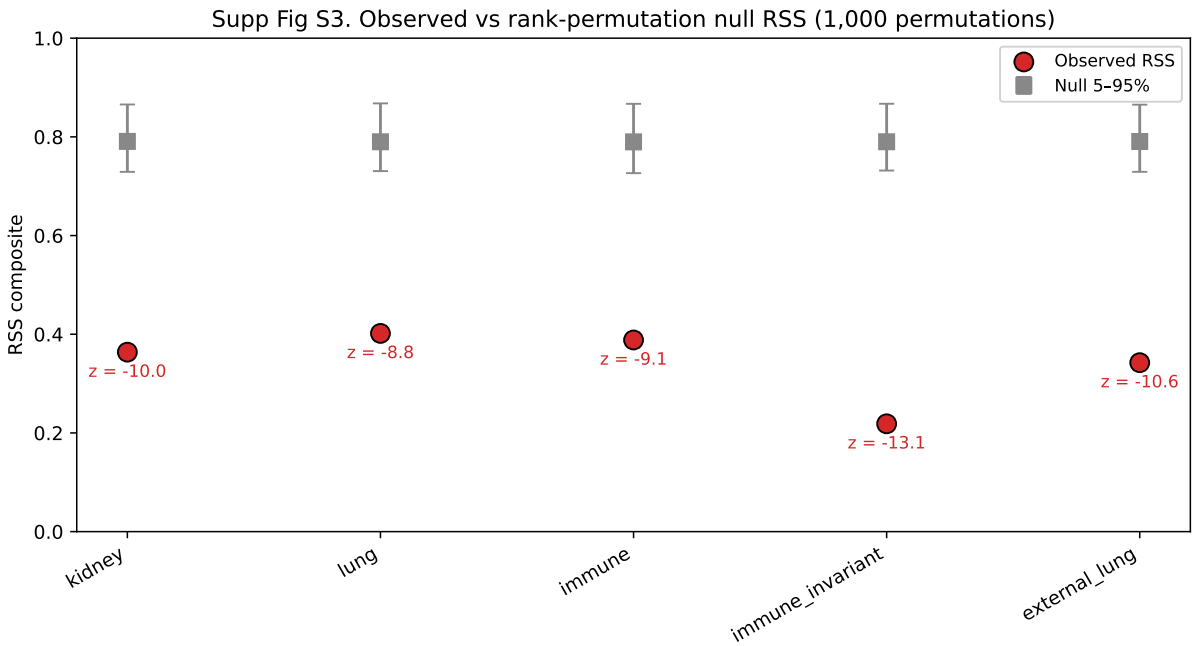

Figure S3: **Empirical null distribution of the bounded-drift RSS under rank permutation.** Histograms show 1,000 permutation draws per cohort; vertical line marks the observed RSS. Every observed value sits 8.8 to 13.1 standard deviations below the null mean (≈ 0.79); empirical $p = 0.001$ (the floor) for every cohort.

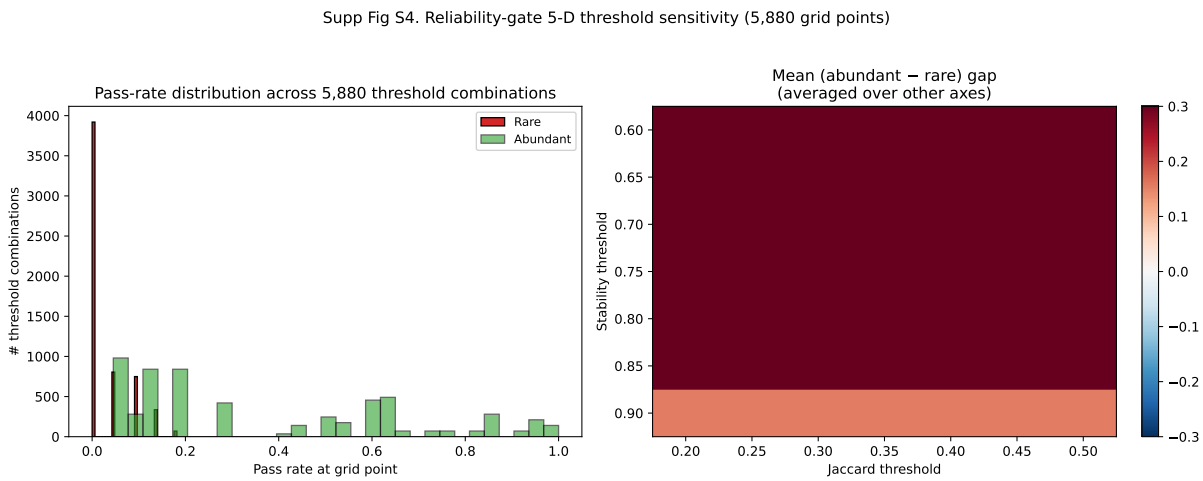

Figure S4: **Reliability-gate threshold-sensitivity heatmap.** Pass-rate gap (abundant minus rare) over the 5,880-point five-dimensional grid (stability, Jaccard, AUC, tail gap, cell count). The gap direction is preserved at 100% of grid points; the primary-threshold gap (+0.182) is conservative relative to the grid mean (+0.344).

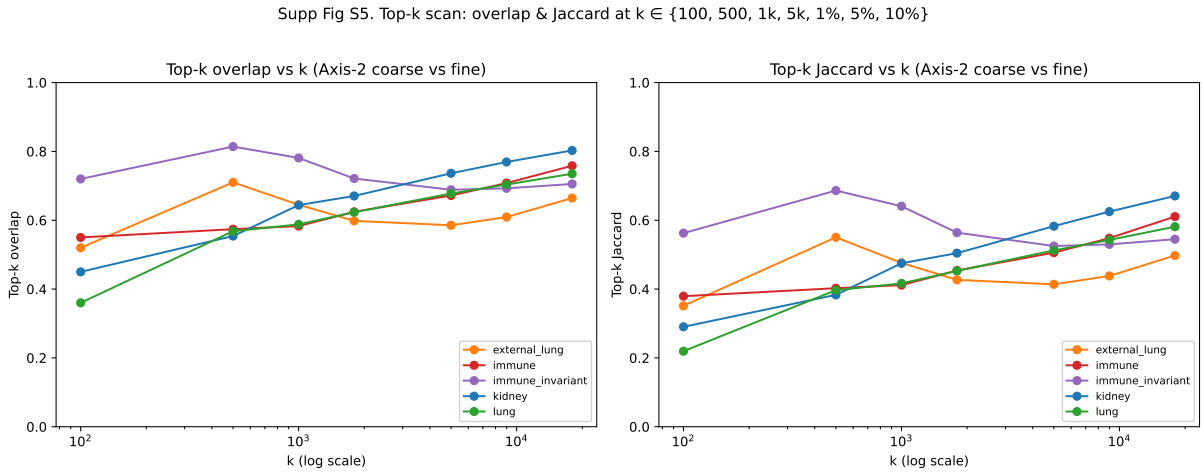

Figure S5: **Top-k scan and per-target overlaps.** Top-k overlap and Jaccard at $k \in \{100, 500, 1{,}000, 5{,}000\}$ plus top-1%, top-5%, and top-10% of the edge universe, and per-target top-5 and top-10 overlaps. Curves remain qualitatively consistent across k, addressing the concern that the headline Jaccard at $k = 1{,}000$ might be artefactual to the chosen k.

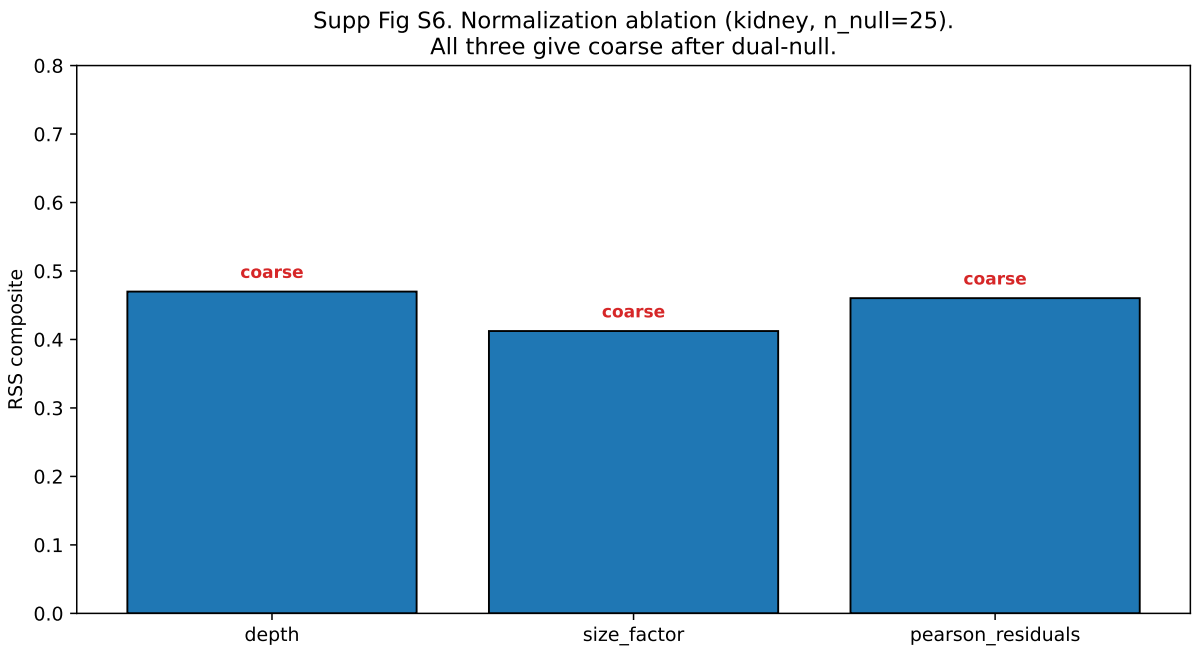

Figure S6: **Normalization ablation on Axis 2 / kidney.** Three schemes compared: depth + log1p (default), median-ratio size-factor + log1p, and analytic Pearson residuals (Lause et al., 2021). All three schemes produce the same calibrated recommendation (*coarse*), rebutting the concern that depth-normalization-induced variance inflation drives the within-coarse failure.

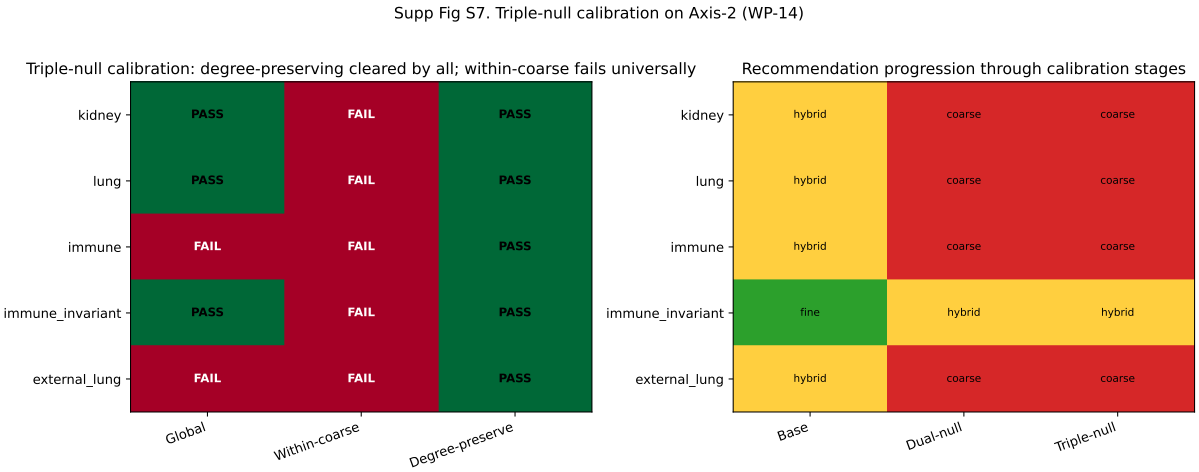

Figure S7: **Triple-null calibration.** A degree-preserving bipartite edge-rewire null is added as a third, deliberately permissive null family. Every cohort clears it at the permutation floor ($p = 0.02$) while continuing to fail the within-coarse constrained shuffle at $p \geq 0.22$, demonstrating that the within-coarse failure reflects genuine over-partitioning of the coarse cell types rather than null over-stringency.